

XDJ-X

All-in-one DJ system

Exchange protocol

Service commands

Assembly name:	Firmware version:
Display module	1.01
Mixer assy	0.41
Deck assy	0.29
Power manager	0.27

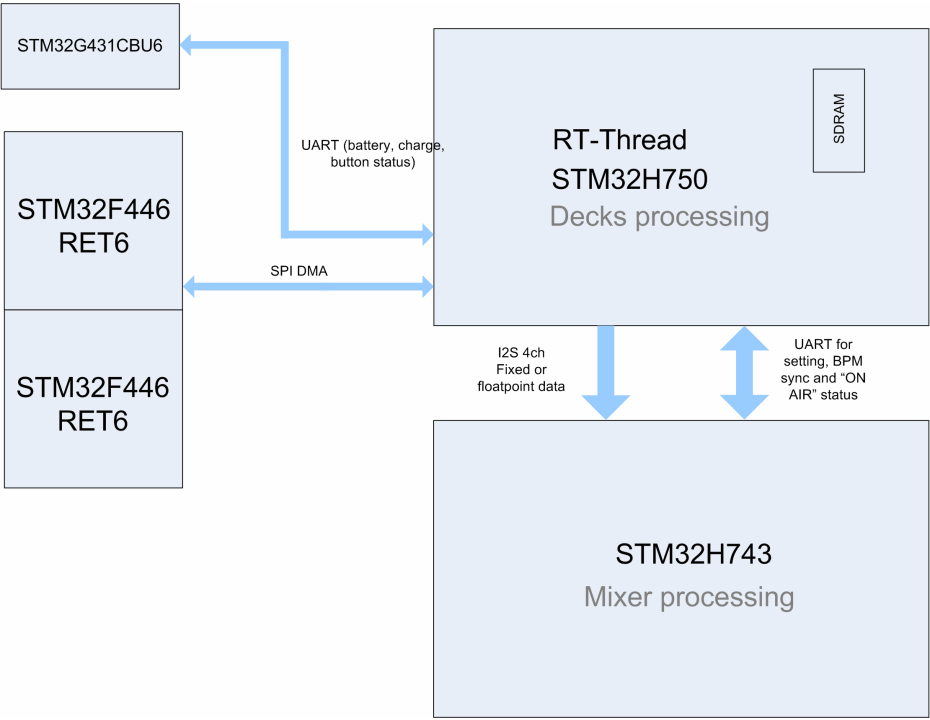
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General connection diagram exchange date:



1. Exchange protocols

Display module <=> Power manager

UART transfer 256000 baud.

Power manager => Display module:

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Bytes	0xAX								0xXX							
Bits	1	0	1	0	0	0	X	X	X	X	X	X	X	X	X	X

Byte[0] - status:

[0xA1] - no AC

[0xA2] - AC charge

[0xA3] - full battery

Byte[1] - data:

[0xAX][0...100] - charge precentage (when the value changes or on request)

[0xAX][0xFE] - button pressed

[0xAX][0xFD] - button unpressed

[0xAX][0xFC] - shutdown command

[0xAX][1xxxxxxx] – firmware version

Display module => Power manager:

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Bytes	0xA1								0xFF							
Bits	1	0	1	0	0	0	0	1	X	X	X	X	X	X	X	X

[0xA1][0x01] - firmware version request

[0xA1][0x55] - battery percentage request

[0xA1][0xFC] - shutdown permission

Display module <=> Mixer Assy

UART transfer 256000 baud.

Display module => Mixer Assy:

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Bytes	0x9X								0xFF							
Bits	1	0	0	1	X	X	X	X	X	X	X	X	X	X	X	X
	10	11	12	13	14	15	16	17								
Bytes	0xFF															
Bits	X	X	X	X	X	X	X	X								

[0x91][0x60][0x31] - firmware version request

[0x92][LOW-MID][MID-HIGH] - setting table values of equalizer cutoff frequencies

[0x93][0xB1][FDR] - setting fader curve type

[0x94][0xA3][HDPH] - setting headphones cue type (pre EQ or post EQ)

[0x95-0x96][BPM][BPM] - sending bpm at beats (BPM*100), 0x95-0x96 – first beat 1/4

[0x97][0xFC][0xFC] - shutdown command

Mixer Assy => Display module:

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Bytes	0x9X								0xFF							
Bits	1	0	0	1	X	X	X	X	X	X	X	X	X	X	X	X
	10	11	12	13	14	15	16	17								
Bytes	0xFF															
Bits	X	X	X	X	X	X	X	X								

[0x91][FW][FW] - firmware version (FW*100)

[0x97][DECKA][DECKB] – on air status

Display module <=> Deck assy

SPI DMA transfer

Deck assy => Display module:

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
Bytes	PADS								BUTTONS							
Bits	PD7	PD6	PD5	PD4	PD3	PD2	PD1	PD0	RLP	LP8	LP4	VNL	SLIP	REV	CUE	PLAY
	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
Bytes	BUTTONS						ADC_PITCH (10 bits)									
Bits	JSNS	TMP	NXT	PREV	RD	0	P	P	P	P	P	P	P	P	P	P
	20	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	2E	2F
Bytes									JOG_SPEED_MSB							
Bits	DIR	FW	FW	FW	FW	FW	FW	FW	X	X	X	X	X	X	X	X
	30	31	32	33	34	35	36	37	38	39	3A	3B	3C	3D	3E	3F
Bytes	JOG_SPEED_LSB								CRC							
Bits	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

ADC_PITCH 0...1023, 512 – center zone

RD – rotation detect

DIR - jog wheel rotation direction

FW – firmware version in code (0-127)

JOG_SPEED 0...65535 16 bits resolution (when jog stopped speed = 0xFFFF)

CRC = (0byte + 1byte + ... + 6byte) % 256

Display module => Deck assy:

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
Bytes	NUM_P								PADn_RED							
Bits	RNGr	RNGg	RNGb	LD1	LD0	nP	nP	nP	R	R	R	R	R	R	R	R
	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
Bytes	PADn_GREEN								PADn_BLUE							
Bits	G	G	G	G	G	G	G	G	B	B	B	B	B	B	B	B
	20	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	2E	2F
Bytes	PLAY_POS								SLIP_POS							
Bits	X	X	X	X	X	X	X	X	0	X	X	X	X	X	X	X
	30	31	32	33	34	35	36	37	38	39	3A	3B	3C	3D	3E	3F
Bytes	CUE_POS								CRC							
Bits	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

nP	LD1	LD0
0	CUE	PLAY
1	SLIP	REV
2	LP4	VNL
3	RLP	LP8
4	TMP	VNLr

5	SNSr	MEMr
6	0	0
7	0	0

nP - package number and at the same time pad number

PADn_(RED, GREEN, BLUE) RGB 888 color for PAD[NUM_P]

RNGr, RNGg, RNGb – RGB bits <<[NUM_P]

MEMr, SNSr, VNLr – VFD segments

SLIP_POS 0...84. 85 – SLIP disable

CUE_POS 0...84. 85 – CUE without cue on jog display

PLAY_POS 0...134 – jog display position cursor 139 = power on animation 138 = eject animation 137 = load in animation 136 = fill circle on display 135 = empty circle.

CRC = (0byte + 1byte + ... + 6byte) % 256

2. Service commands

3. Utility list

Name:	Array location	EEPROM addr	Parameter list
DECKS			
PLAY MODE	[0]	0x0000	SINGLE
			CONTINUE
LOAD LOCK	[1]	0x0001	LOCK
			UNLOCK
AUTO CUE MODE	[2]	0x0002	-36dB
			-42dB
			-48dB
			-54dB
			-60dB
			-66dB
			-72dB
			-78dB
			MEMORY
			FIRST BEAT
QUANTIZE	[3]	0x0003	OFF
			ON
TEMPO RANGE DEFAULT	[4]	0x0004	6%
			10%
			16%
			WIDE
COLOR WAVEFORM	[5]	0x0005	BLUE
			RGB
WAVEFORM POSITION	[6]	0x0006	CENTER LINE
			LEFT LINE
JOG ON AIR	[7]	0x0007	OFF
			ON
JOG COLOR ON AIR	[8]	0x0008	WHITE
			PINK
			RED
			ORANGE
			YELLOW
			GREEN

			AQUA
			BLUE
			PURPLE
			TRACK COLOR
JOG COLOR INACTIVE	[9]	0x0009	WHITE
			PINK
			RED
			ORANGE
			YELLOW
			GREEN
			AQUA
			BLUE
			PURPLE
			TRACK COLOR
JOG BRIGHTNESS	[10]	0x000A	1
			2
LCD BRIGHTNESS	[11]	0x000B	1
			2
			3
			4
			5
PADS MODE	[12]	0x000C	NORMAL
			GATE
MIXER			
FADER CURVE	[13]	0x000D	LINEAR
			SLOW
SPEAKERS	-		OFF
			ON
EQ frequency LOW-MID	[14]	0x000E	
EQ frequency MID -HIGH	[15]	0x000F	
HEADPHONES CUE	[16]	0x0010	PRE EQ
			POST EQ
GENERAL			
BATT. INFO	-		
VERSION No.			
DECK 1	-		0.29
DECK 2	-		0.29
MAIN ASSY	-		1.03
MIXER ASSY			0.41
POWER MANAGER	-		0.27

DEVELOPER	-	ANATSKA ANDREI
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